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International Conference on Intelligent Computing and Sustainable Technologies : Submission (460) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org>

Mon, May 18, 2026 at 8:08 PM

To: gokulpn420@gmail.com

Hello,

The following submission has been edited.

Track Name: IntCST2026

Paper ID: 460

Paper Title: Youth Mental Wellness Platform using Fine-Tuned BERT for Suicidal Ideation Detection

Abstract:

The rising prevalence of mental health disorders among adolescents, coupled with barriers such as stigma, privacy concerns, and lack of accessibility, necessitates scalable digital interventions. This paper presents the design and implementation of a privacy-first mental wellness platform powered by Natural Language Processing (NLP). The core novelty of the system lies in its use of a fine-tuned BERT (Bidirectional Encoder Representations from Transformers) model to detect high-risk language and suicidal ideation with high recall. Unlike generic chatbots, this system integrates an empathetic AI companion, explainable mood analytics via topic modeling, and a tiered crisis escalation workflow.

We evaluate the model against robust traditional machine learning baselines (LinearSVC, Logistic Regression, Naive Bayes, Random Forest), demonstrating that our fine-tuned BERT model achieves a superior F1-score of 0.98, outperforming even the strongest baseline (LinearSVC, F1=0.94). The system architecture prioritizes user privacy through on-device inference capabilities and strict data minimization protocols, offering a secure environment for youth to seek support.

Index Terms—Mental Health, NLP, BERT, Suicide Prevention, Privacy-Preserving AI, Deep Learning

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Primary Subject Area: Data Analytics & Intelligent Applications

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Submission Files:

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Submission Questions Response:

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3. Certificate of originality
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6. Authors Contributions

Data collection and preprocessing: Samar Chhabra

Model training, fine tuning: PN Gokul

7. Conference Rules and Guidelines

Agreement accepted

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